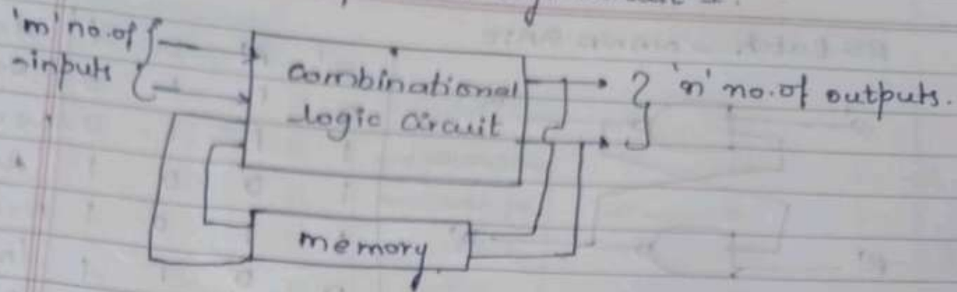
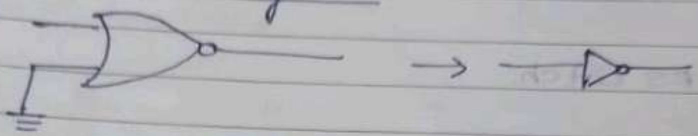
Day-4

Sequential logic Circuit-I.



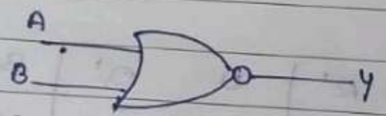
1-bit memory cell.



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

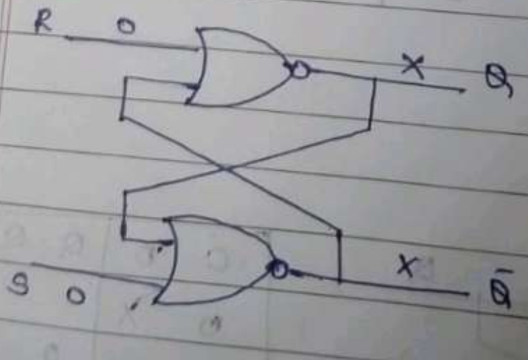
if A=0 , then Y=B'

Behavior of a NOR Gate.



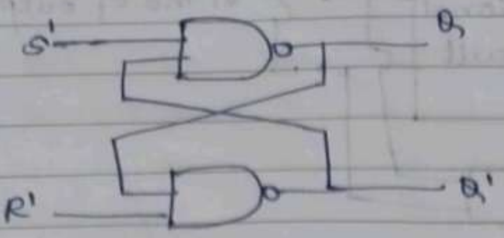
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

RS Latch



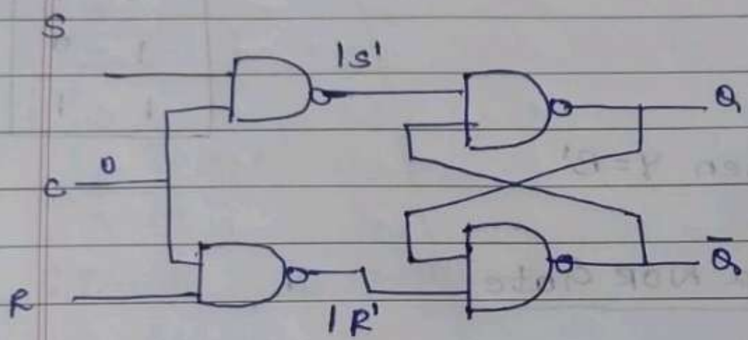
R	S	Q	$\bar{Q}$	State
0	1	1	0	Set
0	0	1	0	No change
1	0	0	1	Reset
0	0	0	1	No change
1	1	0	0	Invalid
0	0	x	x	Race

RS Latch - NAND GATE



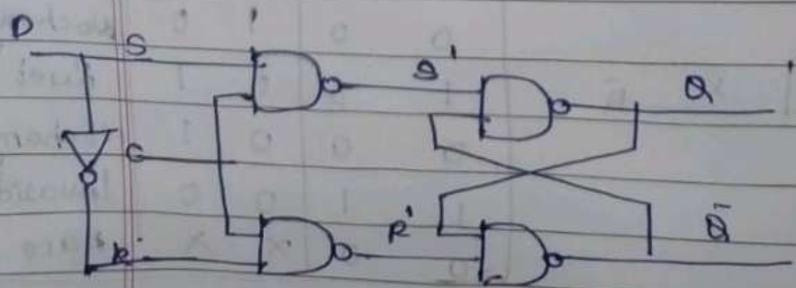
S'	R'	Q	Q'	State
0	1	1	0	Set
1	1	1	0	No cha
1	0	0	1	Reset
1	1	0	1	No cha
0	0	1	1	Invali
1	1	-	-	Race

Gated RS Latch



C	S	R	S'	R'	Q	Q'	State
0	x	x	1	1			No change
1	0	0	1	1			No change
1	0	1	1	0	0	1	Reset
1	1	0	0	1	1	0	Set
1	1	1	0	0	-	-	Invalid

Gated D latch

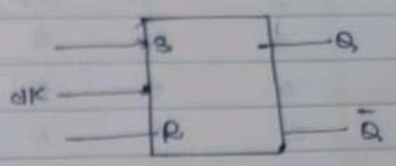


C	D	Q	Q'	State
0	x			No change
1	0	0	1	Reset
1	1	1	0	Set

SR Flipflop

Truth table

S	R	Q	State
X	X	Q	No change
0	0	Q	No change
0	1	0	Reset
1	0	1	set
1	1	-	Invalid

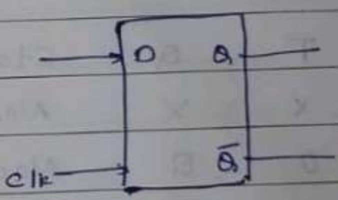


Excitation Table

Q	Q_+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

D flipflop.

clk	D	Q	State.
↓	X	Q	No change
↑	0	0	Reset
	1	1	set

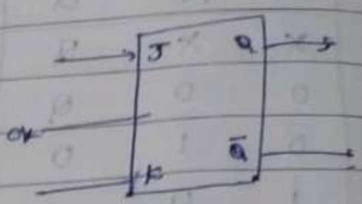


Exitation Table.

Q	Q_+	D.
0	0	0
0	1	1
1	0	0
1	1	1

JK flipflop.

J	K	Q_+	state
X	X	Q	No change
0	0	Q	No change
0	1	0	Reset
1	0	1	Set
1	1	Q'	Toggle

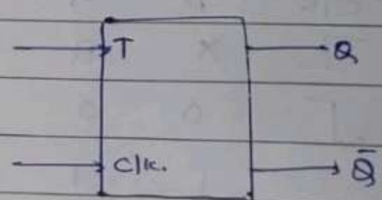


Excitation Table.

Q	Q_+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

T flipflop.

T	Q	State
X	X	Nochange
0	Q	Nochange
1	$Q^{\dagger}$	Toggle



Q	Q_+	T
0	0	0
0	1	1
1	0	1
1	1	0